



PROFILE

ML/AI engineer who builds systems end to end — from geometry and model math through tested APIs, web and mobile clients, and containerised deploys. I keep the hard part pure and provable: cores are unit-tested against inputs with known answers, cameras and models plug in at the edges. Eight open-source systems, 93 automated tests, CI on every repository.

SELECTED PROJECTS

AI Digital Tailor *Python · FastAPI · MediaPipe · Flutter · Docker*

83.8% holdout accuracy on the size-fit classifier

- Estimates body measurements from phone photos: height-calibrated pose landmarks with an elliptical torso cross-section model (Ramanujan perimeter) for chest, waist and hip.
- Real-time guided capture in the browser with live pose-quality gating; burst frames are landmark-averaged and skin tone is gray-world corrected before sampling.
- Parametric 3D avatar exported as OBJ/MTL, virtual try-on preview, and a Flutter mobile client with a pure-Dart, unit-tested capture state machine.

AI Workout Coach *Python · MediaPipe · OpenCV · SQLite · Streamlit*

8 exercises coached in real time; 29 tests covering rep logic

- Hysteresis rep-counting state machines over joint-angle geometry, with per-rep depth scoring and per-frame posture cues (torso lean, hip sag, overhead lockout).
- Workout mode switches exercises live via number keys, persisting each segment as its own session; day-grouped dashboard compares today against yesterday.

Gesture-Based Coding Assistant *Python · MediaPipe Hands · Speech Recognition*

6 gestures and a 9-intent voice grammar compiling to 2 languages

- Rotation-invariant finger-state detection with an edge-triggered, majority-vote gesture stream so a held gesture fires exactly one action.
- Regex command grammar parses spoken instructions into intents, generating Python or JavaScript injected into the focused IDE via keystroke emulation.

Multi-Agent Task Orchestrator *Python · Redis Pub/Sub · FastAPI · SQLite*

5 autonomous agents with planning, monitoring and failure recovery

- Agents decompose, execute and recover tasks over Redis Pub/Sub with SQLite-persisted state and a live Gantt timeline of every step's lifecycle.

Real-Time Stream Engine *Python · Redis Streams · FastAPI · Streamlit*

Dual-tier alerting: threshold plus z-score anomaly detection

- Producer/consumer-group pipeline processing events continuously, routing warnings and critical alerts to dedicated streams with a live monitoring dashboard.

ALSO SHIPPED

Predictive Analytics Platform — ETL to SQLite warehouse, ARIMA training with persisted artifacts, forecast API and dashboard. · Personal Finance Assistant — intent classification and entity extraction routing natural-language messages into a finance database, React chat UI. · Options Strategy Visualizer — clean-architecture Flutter app across six platforms with a live payoff engine and FastAPI/OAuth backend.

TECHNICAL SKILLS

Languages	Python, Dart, JavaScript, SQL
ML / CV	MediaPipe, OpenCV, scikit-learn, NumPy, ARIMA forecasting, 3D geometry
Backend	FastAPI, REST APIs, Redis (Streams & Pub/Sub), SQLAlchemy, SQLite
Frontend / Mobile	React, Tailwind CSS, Vite, Flutter, Streamlit
Engineering	Docker, GitHub Actions CI, pytest, Git, clean architecture

HOW I WORK

"I design systems that stay verifiable under real-time constraints." Pure, dependency-free cores tested against synthetic inputs with known answers. Documented limitations and a stated upgrade path instead of overclaiming. Nothing ships without tests, CI and a README a stranger can follow.

